

Database Schema & Entity-Relationship Specification

1. RELATIONAL DATABASE OVERVIEW (10 NORMALIZED TABLES)

Engine: PostgreSQL / SQLite (SQLAlchemy ORM) | Foreign Keys: Enforced | Composite Indexes: 15 Active

2. ENTITY-RELATIONSHIP TABLE DEFINITIONS & FOREIGN KEYS

- 1. users (PK: id) -> Stores CITIZEN, POLICE, CONSUMER_RIGHTS, and ADMIN accounts + Porichoy NID & Station.
- 2. crime_reports (PK: id, UK: caseId) -> FK: reporterId -> users(id), assignedOfficerId -> users(id).
- 3. consumer_complaints (PK: id, UK: trackingNumber) -> FK: complainantId -> users(id), assignedOfficerId -> users(id).
- 4. shop_reputations (PK: id) -> Tracks merchant compliance score, verified violations, and DNCRP fines.
- 5. barcode_verifications (PK: barcode) -> BSTI & GS1 Bangladesh (894) registry (AUTHENTIC / COUNTERFEIT_FLAGGED).
- 6. sos_requests (PK: id) -> FK: citizenId -> users(id). Supports Store-and-Forward Offline SOS Uplink Sync.
- 7. emergency_alerts (PK: id) -> Public safety broadcasts with geospatial jurisdiction & expiration timestamps.
- 8. case_messages (PK: id) -> FK: senderId -> users(id). 2-way encrypted inquiry chat for GD & DNCRP dockets.
- 9. notifications (PK: id) -> FK: userId -> users(id). Real-time alert & status transition notifications.
- 10. audit_logs (PK: id) -> Immutable National Security audit log recording actor, IP, resource, and timestamp.

3. CARDINALITY & REFERENTIAL INTEGRITY RELATIONSHIPS

- users (1) ----< (0..N) crime_reports [Citizen files Police GDs; Police Officer assigned to investigate]
- users (1) ----< (0..N) consumer_complaints [Citizen files DNCRP disputes; Intake/Investigation/Adjudication]
- users (1) ----< (0..N) sos_requests [Citizen triggers Offline/Live SOS; routed to nearest Police Station]
- crime_reports / consumer_complaints (1) ----< (0..N) case_messages [2-Way Officer-Citizen Inquiry Thread]
- users (1) ----< (0..N) notifications [Real-time status & emergency broadcast delivery]

Note: High-resolution visual ERD diagram is also included in erd_diagram.png & erd_diagram.svg.